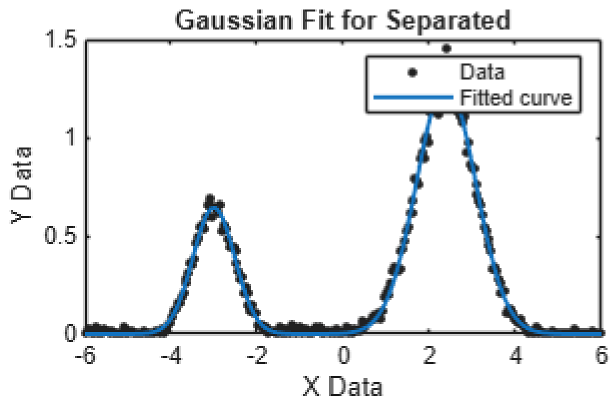
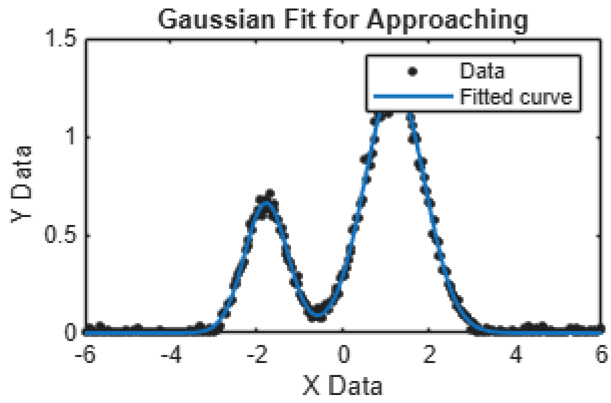
scatterplot 1: separated

```
x1=x1_separated{:,1};
x1=x1(:);
y1=x1_separated{:,2};
y1=y1(:);

ft1 = fittype('gauss2');

[fitresult1, ~] = fit(x1, y1, ft1);

figure;
plot(fitresult1, x1, y1);
xlabel('X Data');
ylabel('Y Data');
title('Gaussian Fit for Separated');
```



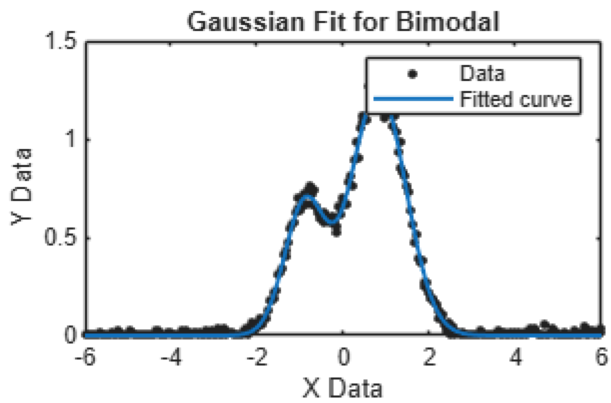
scatterplot 2: approaching

```
x2=x2_approaching{:,1};
y2=x2_approaching{:,2};


ft2 = fittype('gauss2');

% 3. Fit the data
[fitresult2, ~] = fit(x2, y2, ft2);

% 4. Plot the data and the fit
figure;
plot(fitresult2, x2, y2);
xlabel('X Data');
ylabel('Y Data');
title('Gaussian Fit for Approaching');
```



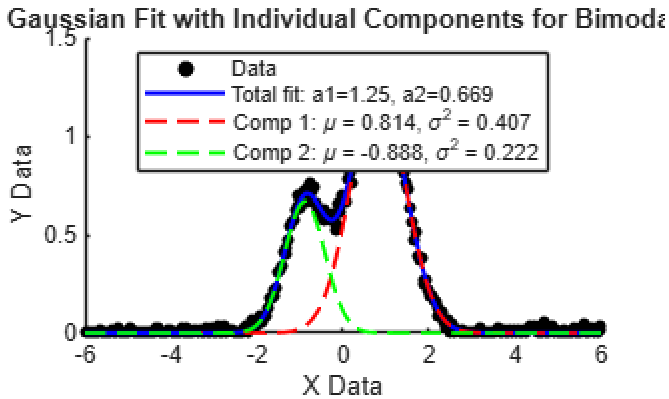
scatterplot 3: bimodal

```
x3=x3_bimodal{:,1};
y3=x3_bimodal{:,2};

ft3 = fittype('gauss2');

% 3. Fit the data
[fitresult3, ~] = fit(x3, y3, ft3);

% 4. Plot the data and the fit
figure;
plot(fitresult3, x3, y3);
xlabel('X Data');
ylabel('Y Data');
title('Gaussian Fit for Bimodal');
```




```
xFine = linspace(min(x3), max(x3), 1000).';

% gaussian coefficients
cv = coeffvalues(fitresult3);
```

```
a1 = cv(1);
b1 = cv(2);
c1 = cv(3);
a2 = cv(4);
b2 = cv(5);
c2 = cv(6);

% gaussian components and total fit
g1 = a1 * exp(-((xFine-b1)./c1).^2);
g2 = a2 * exp(-((xFine - b2)./c2).^2);

gTotal = g1 + g2;

%plot
figure;
hold on
hData = scatter(x3, y3, 20, 'k', 'filled');       % data points
hTotal = plot(xFine, gTotal, 'b-', 'LineWidth', 1.6); % combined fit
h1 = plot(xFine, g1, 'r--', 'LineWidth', 1.4);    % component 1
h2 = plot(xFine, g2, 'g--', 'LineWidth', 1.4);    % component 2
xlabel('X Data');
ylabel('Y Data');
title('Gaussian Fit with Individual Components for Bimodal');
legend('Data', 'Total Fit', 'Component 1', 'Component 2', 'Location', 'Best');
hold off;

% Compute means and variation
mu1 = b1;
mu2 = b2;
sigma1 = c1 / sqrt(2);
sigma2 = c2 / sqrt(2);
var1 = sigma1.^2;
var2 = sigma2.^2;

% Build legend strings with numeric formatting
fmt = '%.3g'; % change format as desired
sData = 'Data';
sTotal = sprintf('Total fit: a1=%.3g, a2=%.3g', a1, a2); % optional
sG1 = sprintf('Comp 1: \\mu = %s, \\sigma^2 = %s', sprintf(fmt, mu1), sprintf(fmt,
var1));
sG2 = sprintf('Comp 2: \\mu = %s, \\sigma^2 = %s', sprintf(fmt, mu2), sprintf(fmt,
var2));

% Use legend entries that include mean and sigma
legend([hData hTotal h1 h2], sData, sTotal, sG1, sG2, 'Location', 'Best');
hold off;
```


Gaussian Fit with Individual Components for Bimoda
Data
Total fit: a1=1.25, a2=0.669
Comp 1: μ = 0.814, σ² = 0.407
Comp 2: μ = -0.888, σ² = 0.222
Y Data
X Data
1.5
1
0.5
0
-6
-4
-2
0
2
4
6